

PERFORMANCE EVALUATION OF GRAPHICS API MIGRATION FROM OPENGL TO DIRECTX IN REAL-TIME VOLCANIC PLUME VISUALIZATION

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Abstract

Effective real-time visualization of dynamic natural hazards like volcanic eruptions plays a crucial role in disaster preparedness and public education, yet traditional scientific simulations often compromise real-time interactivity due to high computational overhead. While existing interactive tools like the OpenGL-based AnitoPlume simulator achieve real-time rendering, their reliance on high-level graphics APIs introduces significant CPU driver overhead and limits explicit control over modern GPU resources. This study addresses the challenge of modernizing scientific visualization systems by migrating the rendering subsystem of AnitoPlume to a low-level explicit graphics API, exploring whether the increased implementation complexity yields practical architectural and performance benefits. The primary contribution of this research is a modernized, DirectX-based rendering framework for the AnitoPlume simulator that preserves original physical behaviors and visual output while exposing direct hardware control. To achieve this, the methodology involves systematically porting the rendering pipeline, shader programs, and memory management structures from OpenGL to DirectX, followed by a comparative performance evaluation under identical hardware configurations. This research establishes an empirical baseline for graphics API migration in scientific visualization, providing a highly scalable foundation for integrating modern hardware-accelerated rendering techniques.

Keywords: AnitoPlume, real-time visualization, DirectX, OpenGL, graphics API migration

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Chapter 1

Introduction

1.1 Background of the Study

Volcanic eruptions are among the most destructive natural hazards, producing ash plumes, pyroclastic flows, and volcanic gases that threaten nearby communities and critical infrastructure. Effective visualization of these hazards plays an important role in disaster preparedness, scientific communication, and public education. Traditionally, volcanic simulations have relied on numerical models that prioritize scientific accuracy for hazard prediction and analysis. While these models provide valuable quantitative information, they are often computationally intensive and lack the interactive capabilities necessary for real-time visualization (Rossant & Rougier, 2021).

To overcome these limitations, the sub-field of real-time computer graphics has increasingly been applied to scientific visualization, balancing visual plausibility with high interactive performance. Within this domain, a notable localized development is AnitoPlume, a real-time three-dimensional volcanic plume simulator designed specifically for visualizing the complex eruption dynamics of Taal Volcano (Del Gallego et al., 2026). Developed using C++ and the OpenGL graphics API, this specialized application combines a Lagrangian particle-based plume system with a detailed digital elevation model of the volcano’s terrain. By maintaining an interactive rendering performance of approximately 30 frames per second, the system successfully demonstrated that computer graphics could bridge the gap between complex geological events and accessible hazard awareness software for local education and planning (Del Gallego et al., 2026).

Previous efforts in graphics-driven volcanic visualization have heavily relied on

traditional higher-level APIs like OpenGL due to their broad driver portability and structural simplicity (Shiraef, 2016). In these legacy architectures, the graphics API simplifies software development by delegating critical hardware management tasks—such as memory allocation and command scheduling—directly to the vendor’s graphics driver. However, as visualization workloads scale, this abstraction inevitably introduces massive CPU driver overhead and restricts developers from optimizing how the hardware executes commands (Bossard, 2019). Consequently, the computer graphics industry has seen a global shift toward modern, low-level explicit graphics APIs like DirectX (Capodiec, Cavicchioli, & Marongiu, 2022), which strip away driver intervention and grant developers granular, direct control over GPU resources, pipeline states, and memory synchronization to maximize hardware efficiency (Ioannidis & Boutsis, 2020).

Despite the theoretical performance advantages offered by modern explicit rendering pipelines, a significant challenge remains in the software modernization of established scientific tools. High-level visualization platforms like the original AnitoPlume simulator are tightly coupled to OpenGL architectures, creating a rigid barrier to optimization and preventing the integration of advanced, contemporary rendering techniques. Transitioning an active simulation system to a low-level framework like DirectX introduces extreme architectural complexity, requiring a complete restructuring of shader programs, command submissions, and memory allocation strategies (Unterguggenberger, Kerbl, & Wimmer, 2022). It remains an open question whether the practical rendering efficiency, resource utilization, and extensibility gains of such a migration justify the steep technical overhead required to port real-time hazard visualization software without altering its underlying scientific behaviors.

1.2 Research Objectives

The primary aim of this study is to modernize the rendering architecture of the AnitoPlume volcanic plume simulator by porting its OpenGL implementation to DirectX and evaluating the impact of the migration on rendering performance, system architecture, and future extensibility.

The specific objectives of this research are as follows:

1. To port the rendering subsystem of the published AnitoPlume simulator from OpenGL to DirectX 12 while strictly preserving its baseline simulation functionality and visual output;
2. To identify and document the technical engineering challenges encountered

during the migration process, focusing on shader translation, rendering pipeline state adaptation, explicit resource management, and command synchronization.

3. To compare the performance of the OpenGL and DirectX 12 implementations under identical hardware environments using empirical metrics, including frame rate (FPS), CPU/GPU resource utilization, and frame rendering times; and
4. To assess the architectural maintainability and extensibility of the new DirectX 12 foundation for integrating future hardware-accelerated rendering techniques into the simulator.

1.3 Scope and Limitations of the Research

This study is bounded by the technical objectives of migrating an existing graphics application to a modern architecture, establishing strict constraints to isolate rendering engine efficiency.

The architectural analysis is limited to comparing the state-machine model of OpenGL with the explicit, low-overhead pipeline model of DirectX 12 within real-time scientific visualization frameworks. This evaluation is strictly constrained to the Windows operating system environment. It excludes alternative modern APIs like Vulkan or Metal because the baseline AnitoPlume application is native to Windows, justifying DirectX 12 as the most appropriate target platform for native hardware acceleration.

The implementation phase is strictly confined to porting the rendering subsystem of the published AnitoPlume simulator from OpenGL to DirectX 12. The scope includes migrating existing graphics resources, translating shader programs into High-Level Shader Language (HLSL), and rewriting command buffers to accurately reproduce the original terrain and particle rendering. This study explicitly excludes any modifications to the core scientific simulation algorithms, Lagrangian physics math, or the terrain mesh generation developed by Del Gallego et al. This boundary ensures that the physical simulation remains a controlled variable, preventing data contamination during the performance comparison.

The documentation of technical challenges is focused entirely on low-level engineering hurdles native to explicit API migration, such as pipeline state object (PSO) compilation and explicit memory synchronization. The study excludes user-facing software usability, user interface design adjustments, or human

participant testing, as application usability was already verified in the baseline study. Instead, data sources are purely technical, derived from automated profiling tools (e.g., PIX or RenderDoc) capturing frame rate (FPS), rendering times, and CPU/GPU utilization during a standardized 180-second simulated eruption sequence. Testing is bounded to standard consumer-grade discrete graphics hardware to reflect the deployment environment of local disaster-response centers.

Finally, the assessment of future extensibility is limited to structural code evaluations and modularity analysis regarding how the new DirectX 12 architecture prepares the simulator to support contemporary rendering pipelines. The actual implementation of advanced features, such as hardware-accelerated ray tracing or compute shaders, is excluded from this study to avoid scope creep while providing a clear architectural roadmap for future researchers.

1.4 Significance of the Research

This study contributes to both computer graphics research and scientific visualization by examining the modernization of an existing real-time simulation system rather than developing an entirely new application. Through the successful migration of AnitoPlume from OpenGL to DirectX, the research provides practical insights into graphics API migration and software modernization for visualization systems.

For researchers in computer graphics, the study offers a detailed case study demonstrating the architectural modifications required when transitioning between two fundamentally different graphics APIs. The comparison between the OpenGL and DirectX implementations contributes empirical evidence regarding rendering performance, implementation complexity, and software maintainability.

For developers of scientific visualization software, the findings may serve as a reference for future projects that require modernization of legacy rendering systems while preserving existing simulation logic. The documented migration process may also assist developers considering similar transitions to lower-level graphics APIs.

For future researchers, this work establishes a modern DirectX-based rendering foundation that may support the integration of advanced graphics techniques, including hardware-accelerated ray tracing, compute shaders, GPU-driven rendering, and other contemporary visualization methods. Consequently, the study extends the long-term applicability of the original AnitoPlume project beyond its initial OpenGL implementation.

Chapter 2

Related Works

This chapter provides a synthesis of past research, existing algorithms, and or state-of-the-art software that are related/similar to the thesis. It should not present detailed summaries of each related work, but rather present a cohesive comparison of different aspects of their work. At the end of each section and this chapter, it should be clear what research challenges and opportunities will be focused on for the proposal.

The sections can be about approaches, application areas, and categories of solutions that give readers an deep understanding of the current state of the field.

Observe a consistent format when presenting each of the reviewed works. This must be selected in consultation with the prospective adviser.

DO NOT FORGET to cite your references. Related works can be discussed multiple times in different sections of this chapter, depending on what is being discussed or compared.

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